Sri Lanka Institute of Information Technology

Web-based Help Desk for University Students

MLB-B2G2-10	
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Software Engineering | SE2030

B.Sc. (Hons) in Information Technology

1. Sprint summaries

Sprint 1: Core Functionality

Goal: Implement authentication, ticket submission, and basic dashboards.
User Stories Completed:

- PBI-01: Student ticket submission with attachments
- PBI-06: User role management
- PBI-07: Real-time ticket tracking
- PBI-05: IT ticket resolution basics

Outcome:

- Users can log in and submit tickets.
- Basic dashboards for Students and IT Officers are functional.
- Role-based access control is implemented.

Sprint 2: Workflow Management

Goal: Enable ticket assignment, status updates, and redirection.
User Stories Completed:

- PBI-02: Ticket assignment to team members
- PBI-11: Redirect misrouted tickets
- PBI-03: Lecturer reply functionality
- PBI-12: Assign tickets to lecturers

Outcome:

- Staff can assign and redirect tickets.
- Lecturers can reply to and resolve academic tickets.
- Ticket statuses are now updatable (e.g., In Progress, Resolved).

Sprint 3: Self-Service & Analytics

Goal: Launch knowledge base, feedback, and reporting features.
User Stories Completed:

- PBI-09: Searchable knowledge base
- PBI-14: Student rating and feedback system
- PBI-04: Resolution-time reports for Academic Coordinators
- PBI-10: System settings configuration

Outcome:

- Students can search the knowledge base before submitting tickets.
- Feedback and ratings are collected upon ticket resolution.
- Administrators can generate basic analytical reports.

Sprint 4: Optimization & Admin Tools

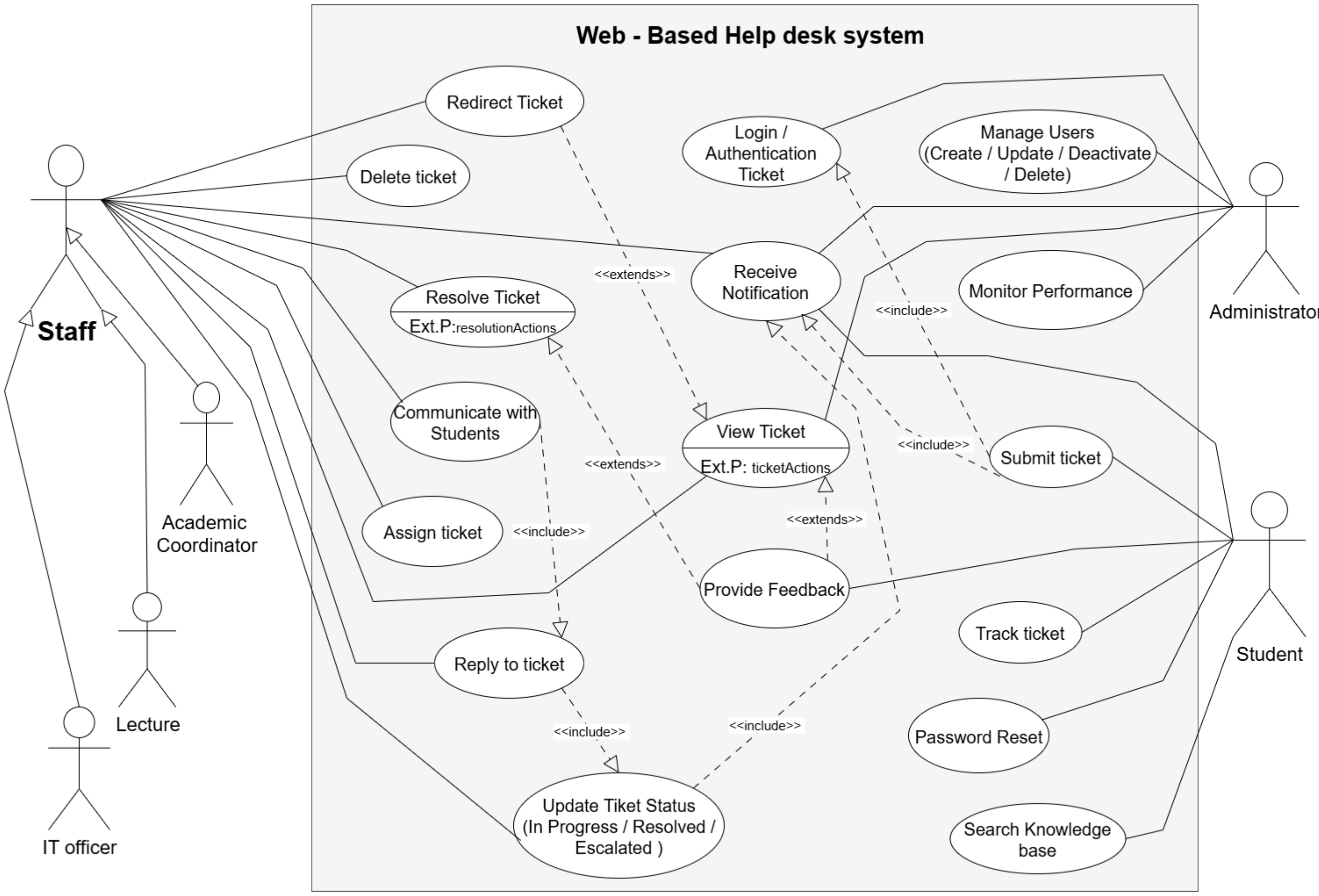
Goal: Enhance security, diagnostics, and user management.
User Stories Completed:

- PBI-08: Escalation of complex issues
- PBI-13: Reassign misrouted tickets
- PBI-15: Deactivate outdated user accounts
- PBI-16: Diagnostic tools for IT Officers

Outcome:

- Improved ticket escalation and reassignment workflows.
- Admins can deactivate users and manage accounts securely.
- IT Officers have access to diagnostic tools for faster resolution.

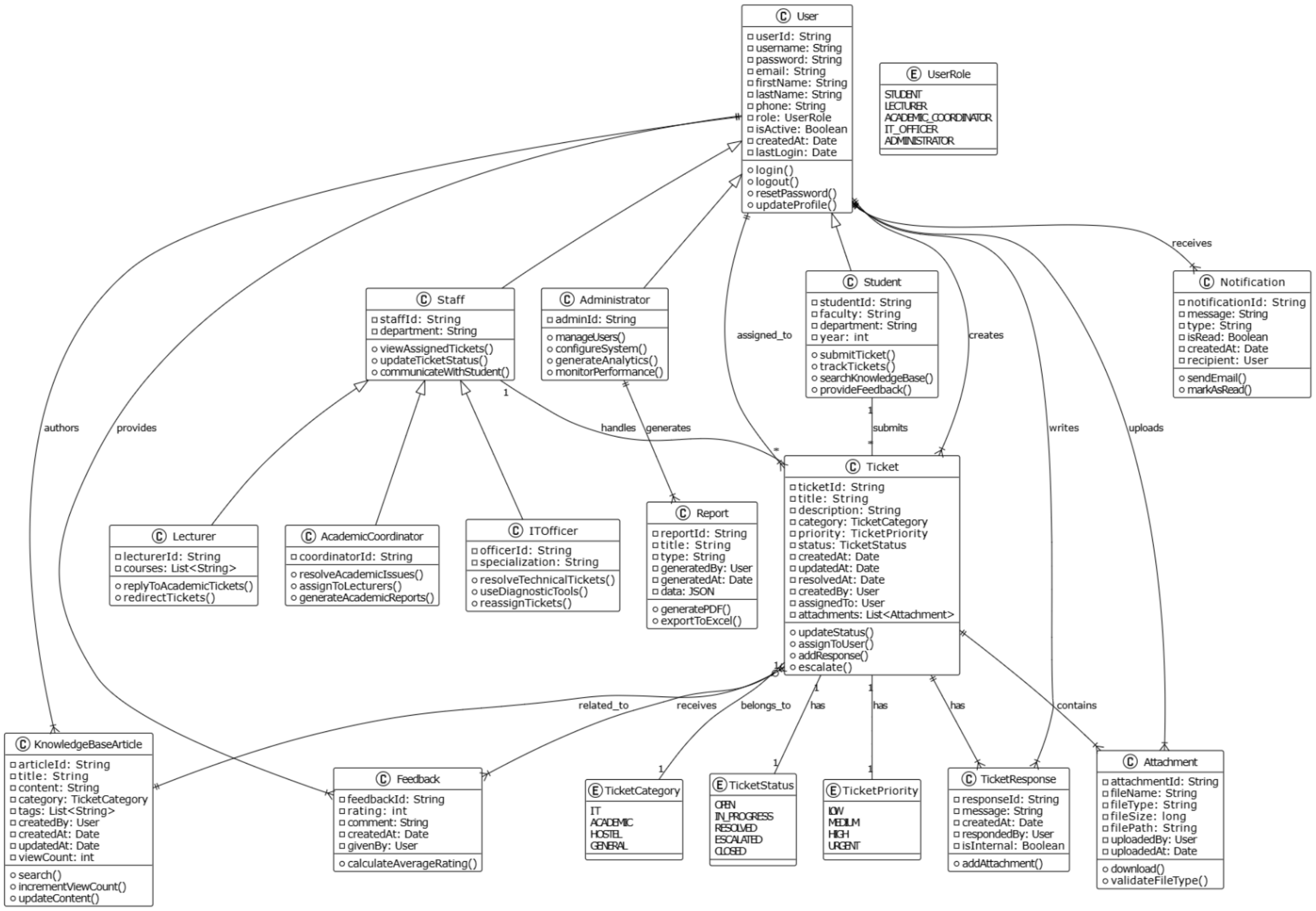
2. Use Case Diagram



View Ticket *Ext. Point* → "ticketActions" (covers Reply to ticket, Assign/Redirect, Update Ticket Status, Delete, Provide Feedback, Notify)

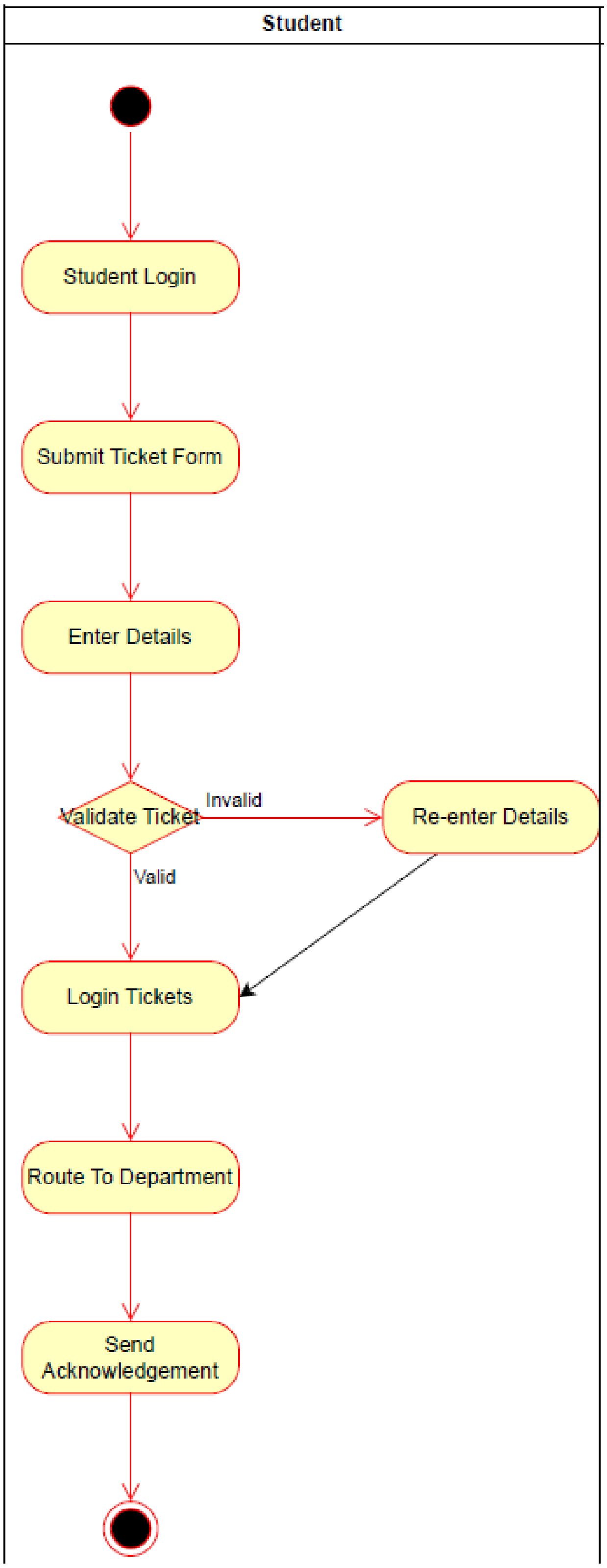
Resolve Ticket *Ext. Point* → "resolutionActions" (covers Reassign/Redirect, Escalate, Update Status to Resolved, Notify, Ask for more info)

3. Class Diagram

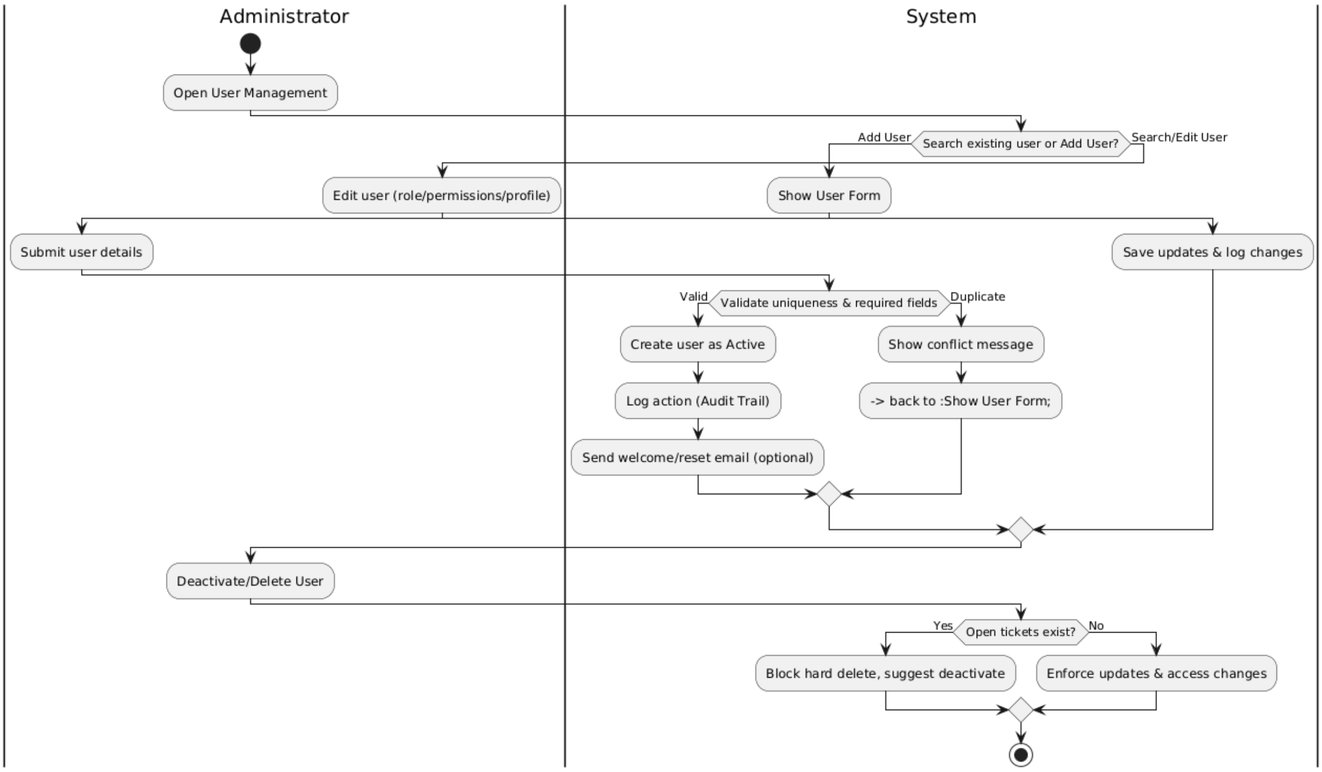


4. Activity Diagrams

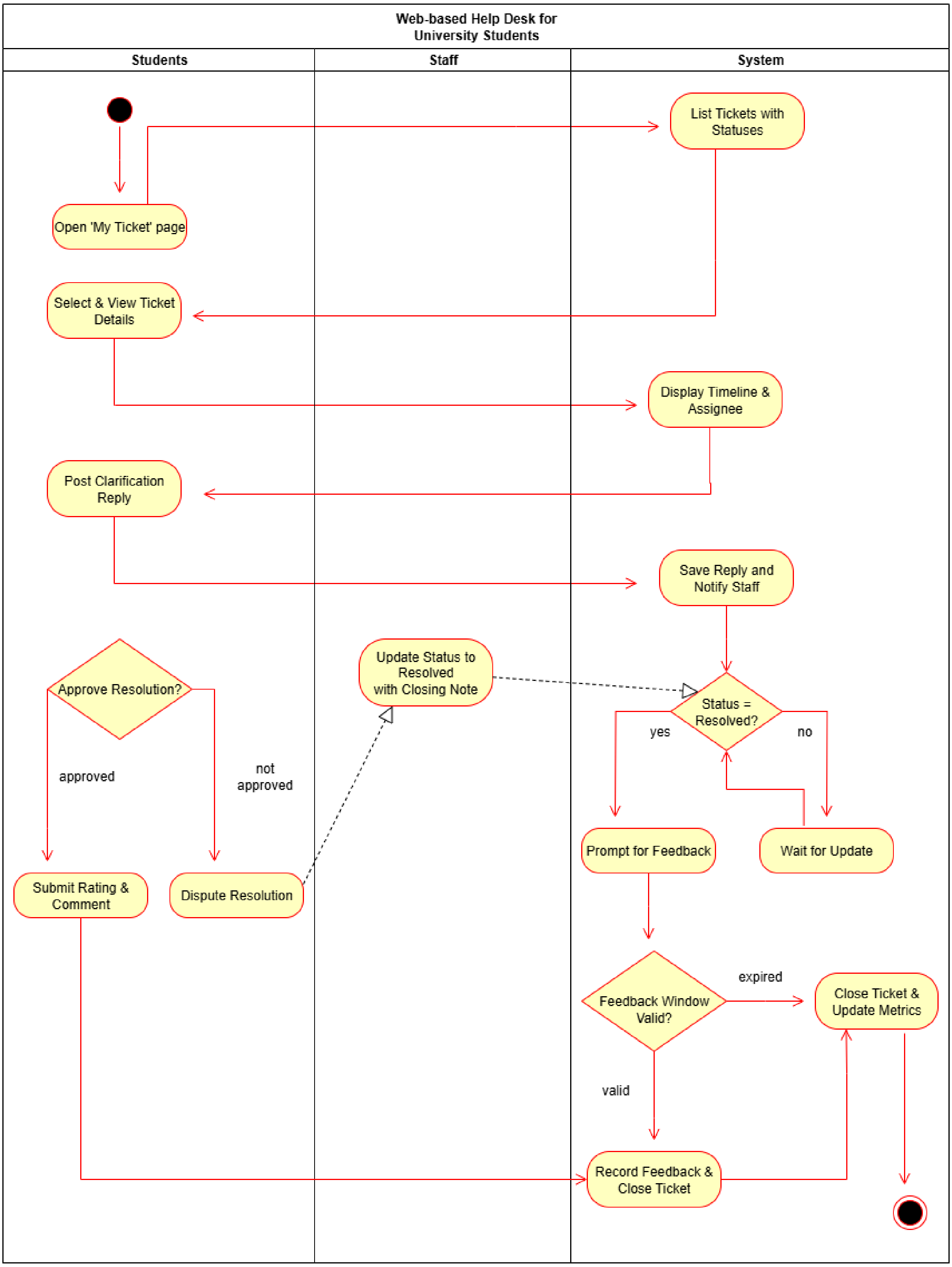
1. Rosa S.H.V.A — Ticket Management



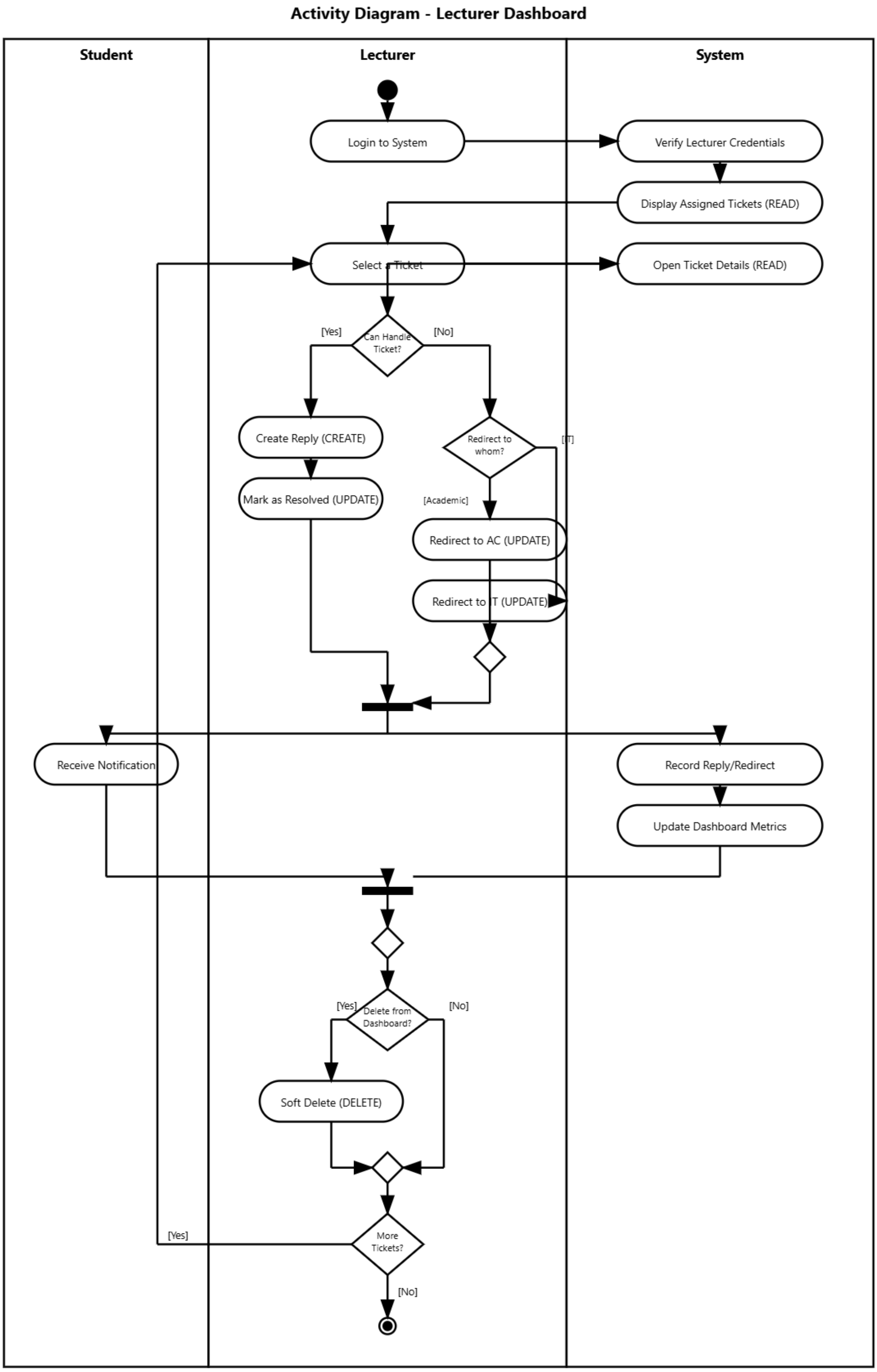
2. Ruth R W N — User Management



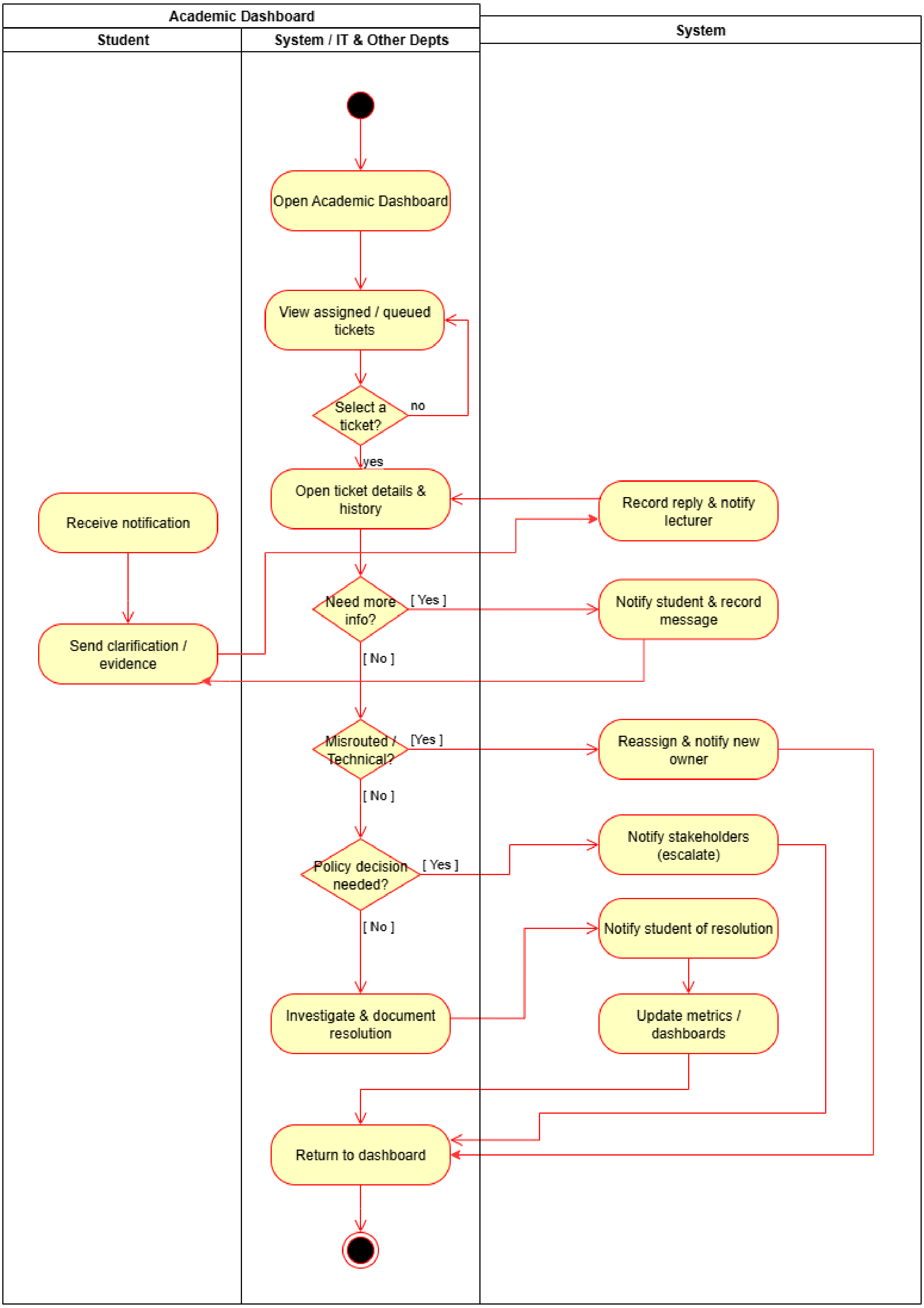
3. Ilesingha I.Y.R — Student Dashboard



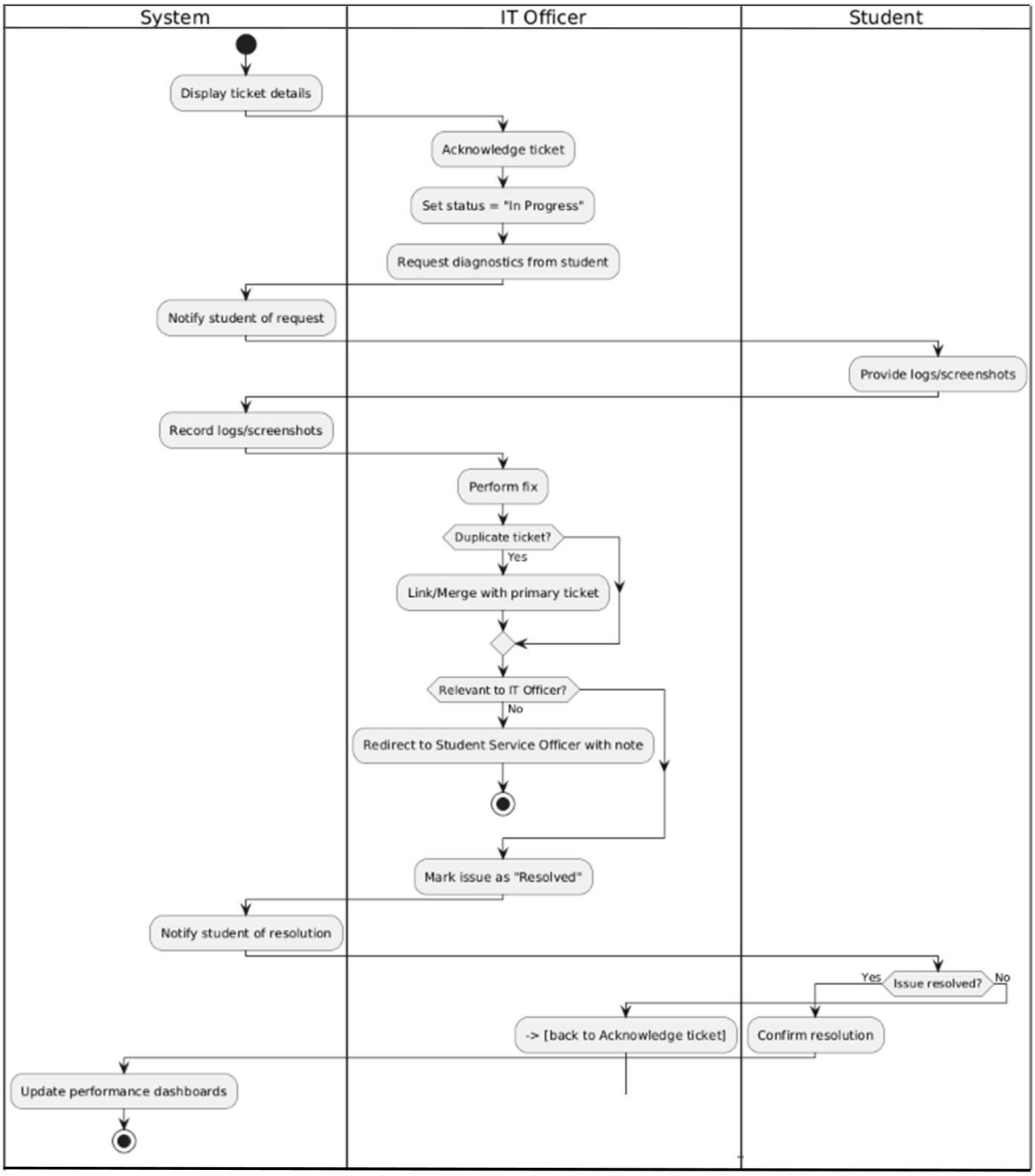
4. Hanwellage P.M — Lecture Dashboard



5. Pinthu D.I.U — Academic Dashboard



6. Yeshan K.P — IT Officer Dashboard



5. Ethical Considerations

Our system processes sensitive student and staff information, provides the most important university services, and it can make a significant difference in the academic life of users. Consequently, the most important thing is to stick to the principles of ethics.

i. Privacy and Data Confidentiality

- **Consideration:** The system gathers and stores personal data (names, IDs, emails) and even sensitive information (academic problems, IT problems, hostel maintenance requests).
- **Implications:** Hacker attacks or unauthorized access to data may result in the embarrassment, discrimination, or abuse of personal data.
- **Mitigation Strategies:**
  - The data stored at rest and in transit (HTTPS) should be encrypted with strong encryption (AES-256) as your non-functional requirements imply.
  - Follow the least privilege principle in your role based access control. As an example, a lecturer should not be able to see all academic tickets, just those associated with his/her courses.
  - Establish specific data retention policies. What is the life span of resolved tickets and other related personal information? They are not supposed to be held forever.
  - Include data in reports to administrators in an anonymized form in order to avoid identification.

ii. Data Security and Integrity

- **Consideration:** his system is a center of reporting concerns. Stolen would be exploited to enter fraudulent tickets, destroy valid ones or compromise their status, which would affect the operations of the university.
- **Implications:** Mistrust in the system, which makes the users go back to the old, ineffective approaches. The possibility of harassment of personnel or students by malicious individuals.
- **Mitigation Strategies:**
  - Use secure coding practices to implement strong authentication (OAuth 2.0) and prevent typical web security threats (SQL injection, XSS).
  - Audit trails should be kept of the key activities (e.g., "User X deleted ticket Y at time Z). This plays a key role in accountability.
  - Automated backups (depending on your level of reliability) to avoid the loss of data and guarantee business continuity.

iii. Fairness, Bias, and Non-Discrimination

- **Consideration:** The auto-routing (e.g. routing a ticket to IT, Academic Coordinator or Lecturer) should be fair and precise.
- **Implications:** Misrouted ticket based on a prejudiced or faulty algorithm may result in a huge delay in the process which will hurt an academic performance of a student.
- **Mitigation Strategies:**
  - Make sure that the auto-routing logic is founded on transparent, clear and fair rules and are reviewed periodically.
  - Offer a user-friendly and straightforward system in which staff (Lecturers, ACs, IT) can reassign misrouted tickets fast as defined in your use cases.
  - Make the system as user friendly as possible (accessible to all users, e.g. compatible with screen readers, navigation by the keyboard and so on) which is both ethically correct and legally required in many cases.

iv. Openness and Responsibility

- **Consideration:** It is important that the users should be aware of how the system operates and the ownership of their data and their tickets..
- **Implications:** A black box system may be frustrating and give the sense of powerlessness to the users who do not know why a ticket was rerouted to a particular destination or why it was closed.
- **Mitigation Strategies:**
  - Post status updates and timelines to the students on their dashboard.
  - Make sure that everything it does (changes in status, reassignments etc.) is registered with the staff member ID.
  - Clear, published privacy policy and terms of use, which describe what the data is collected and used..

v. User Autonomy and Consent

- **Consideration:** Students and staff must be able to manage their data within the confines of the purpose of the system.
- **Implications:** Compelling users to share their data without a need to do so or sharing their data to unknown purposes infringes their autonomy.
- **Mitigation Strategies:**
  - Only gather data which is required to resolve a ticket.

- Give the students the capability of deleting their own tickets in case they are unresolved and not yet responded (according to your CRUD operations), and thus retain the control over erroneous submissions
- Should feedback rating be released publicly (e.g. to demonstrate performance measurements on personnel), make this clear at the outset.

6. Professional and Social Responsibility

- **Consideration:** The development team has a role to play to develop a system that is dependable, and cannot hurt its users or the university at large.
- **Implications:** A bad system: A bad system may increase the issues that a system is meant to address, which creates a waste of resources and time.
- **Mitigation Strategies:**
  - Also follow a strict testing procedure (UAT, black-box, white-box) as scheduled to achieve a high quality and reliability.
  - Assess and adequately train and document staff (an important assumption in your constraints) to make them capable of working with the system in a manner that is both ethical and effective.
  - Provide avenues of escalation to users who may have a complaint about the system itself or the way your ticket was handled.